

EV Battery Management System

Data Analysis Report

Vehicle: EV001 | Date: 2026-07-21 | Generated: 2026-07-22 07:19

1. Executive Summary

This report presents the battery analytics for vehicle **EV001** recorded on **2026-07-21**. A total of **1,729** data points were captured at 50-second intervals across three operating modes: Driving, Charging, and Parking. The analysis covers State of Charge (SOC), State of Health (SOH), voltage, current, temperature, speed, and power metrics.

The vehicle spent **68.2%** of the day parked, **22.4%** driving, and **9.4%** charging. Two charging sessions were recorded during the day.

2. Key Metrics

Metric	Value
Average SOC (%)	80.24
Average SOH (%)	96.5
Average Voltage (V)	385.9
Average Temperature (°C)	30.29
Maximum Temperature (°C)	34.2
Minimum SOC (%)	73.01
Driving Records	387
Charging Records	162
Parking Records	1180

Total Records	1729
Driving Duration (min)	322.5
Charging Duration (min)	135.0
Parking Duration (min)	983.3

3. Mode-wise Breakdown

The table below shows per-mode statistics including record count, duration, average SOC, temperature, and power.

Mode	Records	Duration (min)	Avg SOC (%)	Avg Temp (°C)	Avg Power (kW)
Driving	387	322.5	78.5	30.7	34.61
Charging	162	135.0	77.2	32.7	26.08
Parking	1180	983.3	81.2	29.8	0.09

4. Driving Analysis

During driving, the SOC dropped from **84.3%** to **79.7%**. The average vehicle speed was **47.3 km/h** with a peak of **79.9 km/h**. Average power consumption was **34.61 kW**. Battery temperature during driving ranged from **27.9 °C** to **33.9 °C**.

5. Charging Analysis

Two charging sessions were recorded. SOC increased from **73.1%** to **80.8%** across both sessions combined. Average charging current was **67.5 A**. Battery temperature during charging reached a maximum of **34.2 °C**, which is within safe limits.

6. Parking Analysis

During parking, the SOC remained stable with minimal self-discharge. SOC ranged between **75.1%** and **85.0%**. The battery temperature gradually decreased toward ambient, settling around **29.7 °C**. Current was approximately zero, confirming no parasitic drain.

7. Battery Health Summary

The State of Health (SOH) averaged **96.50%**, indicating the battery pack is in excellent condition. The maximum recorded temperature was **34.2 °C**, well below the critical threshold of 55 °C. No thermal events were detected.